Talk 0: Algebraic K-theory

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1 Algebraic K-theory

Motivation: Euler characteristics (André Weil)

Let M be a compact complex manifold and V a holomorphic vector bundle on M.[1] There are two basic invariants of V:[2]

$$\begin{array}{ccc} \{\text{complex vector bundles on } M\} & \xrightarrow{\text{Chern}} & H^*(M,\mathbb{C}) \\ {\scriptstyle \chi(-)}\big\downarrow & & \\ \mathbb{Z} & & \end{array}$$

Write $\mathrm{triv}_{\mathbb{C}} = M \times \mathbb{C}$ for the trivial line bundle, so that $\mathrm{Hom}_M(\mathrm{triv}_{\mathbb{C}}, V) = \Gamma(V)$. A short exact sequence $0 \to V \to W \to W/V \to 0$ gives a long exact sequence

$$\begin{aligned} 0 \to \Gamma(V) \to \Gamma(W) \to \Gamma(W/V) \to \mathrm{Ext}^1_M(\mathrm{triv}_{\mathbb{C}}, V) \\ \to \mathrm{Ext}^1_M(\mathrm{triv}_{\mathbb{C}}, W) \to \mathrm{Ext}^1_M(\mathrm{triv}_{\mathbb{C}}, W/V) \to \mathrm{Ext}^2_M(\mathrm{triv}_{\mathbb{C}}, V) \to \cdots \end{aligned}$$

and the Euler characteristic is

$$V \longmapsto \bigoplus_n \mathrm{Ext}^n_M(\mathrm{triv}_{\mathbb{C}}, V) \overset{\chi}{\rightsquigarrow} \chi(V) = \sum_n (-1)^n \dim_{\mathbb{C}} \mathrm{Ext}^n_M(\mathrm{triv}_{\mathbb{C}}, V).$$

Remark. $\mathrm{Ext}^n_M(\mathrm{triv}_{\mathbb{C}}, V) \cong H^n(M; V)$ is sheaf cohomology. It is finite-dimensional because M is compact (Cartan–Serre) and vanishes for $n > \dim_{\mathbb{C}} M$, so $\chi(V)$ is defined. Since the alternating sum of dimensions along a finite exact sequence is zero, the long exact sequence gives

$$\chi(W) = \chi(V) + \chi(W/V).$$

Definition.

$$K_0(M) := \mathbb{Z}\{\text{iso. classes of vector bundles on } M\} \big/ \big([W] = [V] + [W/V]\big),$$

one relation for every short exact sequence $0 \to V \to W \to W/V \to 0$.

By additivity, χ factors through $K_0(M)$:

$$\begin{array}{ccc} \mathbb{Z}\{\text{complex vector bundles}/M\} & \xrightarrow{\chi} & \mathbb{Z} \\ & \searrow & \uparrow \\ & & K_0(M) \end{array}$$

So $K_0(M)$ is the universal receptacle for "Euler characteristic-type functions".

[1] For smooth bundles every short exact sequence splits and $\Gamma(V)$ is infinite-dimensional, so the story below needs the holomorphic setting.

[2] They are related by the Hirzebruch–Riemann–Roch theorem $\chi(V) = \int_M \mathrm{ch}(V)\ \mathrm{td}(T_M)$, so χ is determined by the Chern character. Grothendieck introduced K_0 to generalise this to proper maps $f\colon X \to Y$ (Grothendieck–Riemann–Roch), where $\sum_i (-1)^i R^i f_* V$ only makes sense in $K_0(Y)$. See [3, §0, Thm. 0.3, Def. 0.4, Thm. 0.5].

Atiyah–Hirzebruch

Now let X be a compact Hausdorff space and consider (continuous) complex vector bundles on X. Here every short exact sequence splits, so $K(X) = K^0(X)$ is simply the group completion of the monoid of isomorphism classes of vector bundles under $\oplus$.

Notice that $K(-)$ can actually be inflated into a *cohomology theory*. First, it is a contravariant functor: a map $f\colon X \to Y$ pulls back a bundle $W \to Y$ to $f^*W = X \times_Y W$,

$$\begin{array}{ccc} f^*W & \longrightarrow & W \\ \downarrow & \lrcorner & \downarrow \\ X & \xrightarrow{f} & Y \end{array} \qquad\qquad \begin{array}{ccc} \{\mathrm{Vect}\,/Y\} & \xrightarrow{f^*} & \{\mathrm{Vect}\,/X\} \\ \downarrow\!\!\downarrow & & \downarrow\!\!\downarrow \\ K(Y) & \xrightarrow{f^*} & K(X) \end{array}$$

and homotopic maps induce isomorphic pullbacks, so K is homotopy invariant. Then

$$\rightsquigarrow \quad KU^{-*}(X) = \{\text{vector bundles on } \Sigma^* X\}.$$

It is represented by a spectrum, $\mathbb{Z} \times BU \simeq \mathbb{Z} \times BGL(\mathbb{C})$.

Remark. Here $\mathrm{GL}(\mathbb{C})$ carries its usual topology. If one gives $\mathrm{GL}(\mathbb{C})$ the *discrete* topology, $BGL(\mathbb{C})$ becomes a very different space; algebraic K-theory starts from that one (the plus construction $BGL(R)^+$, later in the talk).

K_0, K_1, K_2

$K_0(X)$ has some definition for X a variety over anything.[3] There are more localization sequences, e.g.

$$\bigoplus_{p \in \mathrm{Spec}\,\mathbb{Z} \text{ closed}} K_0(\mathbb{F}_p) \longrightarrow K_0(\mathbb{Z}) \longrightarrow K_0(\mathbb{Q}) \longrightarrow 0,$$

where every term is $\mathbb{Z}$. This is *not exact* on the left!

Next,

$$K_1(R) := \mathrm{GL}(R)/E(R),$$

where $E(R) \subseteq \mathrm{GL}(R)$ is the subgroup generated by elementary matrices (identity plus one off-diagonal entry).

Remark. In the sequence above, the second map is the rank, an isomorphism $\mathbb{Z} \to \mathbb{Z}$. The first map sends $\mathbb{F}_p$ to its class as a $\mathbb{Z}$-module, computed by the resolution $0 \to \mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \mathbb{F}_p \to 0$; so $[\mathbb{F}_p] = [\mathbb{Z}] - [\mathbb{Z}] = 0$ and the first map is zero. Hence the sequence cannot start with $0 \to$; it continues to the left with K_1:

$$K_1(\mathbb{Z}) \to K_1(\mathbb{Q}) \to \bigoplus_p K_0(\mathbb{F}_p) \to K_0(\mathbb{Z}) \to K_0(\mathbb{Q}) \to 0,$$

where $K_1(\mathbb{Q}) = \mathbb{Q}^\times \to \bigoplus_p \mathbb{Z}$ is given by the p-adic valuations (surjective, with kernel $\{\pm 1\} = K_1(\mathbb{Z})$).

[3] As for $K_0(M)$ above, $K_0(X)$ is generated by (algebraic) vector bundles on X modulo short exact sequences. For $X = \mathrm{Spec}\,R$ vector bundles are finitely generated projective R-modules, every short exact sequence of them splits, and $K_0(R)$ is the group completion of $\big(\{\text{f.g. projective } R\text{-modules}\}/\cong,\ \oplus\big)$.

If R is a field, a Euclidean domain or a local ring, then $K_1(R) \cong \mathrm{GL}(R)/\mathrm{SL}(R) \cong R^\times$, via the determinant.

If we have an automorphism of a short exact sequence

$$\begin{array}{ccccccccc} 0 & \longrightarrow & P & \longrightarrow & P' & \longrightarrow & P'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha''} & & \\ 0 & \longrightarrow & P & \longrightarrow & P' & \longrightarrow & P'' & \longrightarrow & 0 \end{array}$$

then the relations $(P,[\alpha])+(P'',[\alpha''])=(P',[\alpha'])$ and $(P,[\alpha\beta])=(P,[\alpha])+(P,[\beta])$ give an equivalent definition of K_1.[4]

Finally, for a field F,

$$K_2(F) = F^\times \otimes_{\mathbb{Z}} F^\times \big/ \langle a \otimes (1-a) : a \neq 0,1 \rangle$$

universally extends the long exact sequence for MV-type situations.[5]

These are the invariants of the field which control the extensions:

$$\begin{array}{l} K_2(F) \xrightarrow{\partial} \bigoplus_{\mathfrak{p}} K_1(\mathcal{O}_F/\mathfrak{p}) \longrightarrow K_1(\mathcal{O}_F) \longrightarrow K_1(F) \xrightarrow{\partial} \\ \qquad \bigoplus_{\mathfrak{p}} K_0(\mathcal{O}_F/\mathfrak{p}) \longrightarrow K_0(\mathcal{O}_F) \longrightarrow K_0(F) \longrightarrow 0 \end{array}$$

with $K_1(F) \cong F^\times$. Here $\mathcal{O}_F$ is a Dedekind domain with fraction field F (e.g. the ring of integers of a number field), $\mathfrak{p}$ runs over the nonzero prime ideals, $K_1(\mathcal{O}_F/\mathfrak{p}) = (\mathcal{O}_F/\mathfrak{p})^\times$, and $\partial\colon K_2(F) \to \bigoplus_{\mathfrak{p}} (\mathcal{O}_F/\mathfrak{p})^\times$ is given by the tame symbols [10, Thm. III.6.5].

1968: Swan, K-V (Karoubi–Villamayor), Segal

Segal: view K-theory as some sort of "derived group completion".[6] The monoid of isomorphism classes of f.g. projective modules under $\oplus$ maps to $K_0(R)$ by a monoid homomorphism,

$$\Big(\{\text{iso. classes of f.g. projective } R\text{-modules}\},\ \oplus\Big) \xrightarrow{\text{monoid hom}} K_0(R),$$

and $K_0(R)$ is the *group completion* of the monoid.

For K_1:

$$K_1(R) = \mathrm{GL}(R)/E(R) \cong \mathrm{GL}(R)^{ab}, \qquad E(R) = [\mathrm{GL}(R), \mathrm{GL}(R)] \text{ is perfect.}$$

To go further, Quillen used a construction on spaces called the *plus construction*. The plus construction $BGL(R) \to BGL(R)^+$ kills off the largest perfect normal subgroup of $\mathrm{GL}(R)$, i.e. $E(R)$: so $\pi_1 BGL(R)^+ = \mathrm{GL}(R)/E(R) = K_1(R)$, while the map induces an isomorphism on homology. Quillen defined $K_n(R) := \pi_n BGL(R)^+$ for $n \geq 1$.

For example, $K(\mathbb{F}_p) \simeq \mathbb{Z} \times BGL(\mathbb{F}_p)^+$, and

$$H^*(BGL(\mathbb{F}_p)^+, \mathbb{Z}/\ell) \cong H^*(\mathrm{GL}(\mathbb{F}_p), \mathbb{Z}/\ell)$$

[4] Bass; cf. [10, Cor. III.1.6.3].

[5] For an arbitrary ring R, $K_2(R)$ can be defined via the Steinberg group $\mathrm{St}(R)$; see [10, Def. III.5.2] and [4, §5]. The formula for fields is Matsumoto's theorem [10, Thm. III.6.1].

[6] Swan, Karoubi–Villamayor and others proposed several (non-equivalent) definitions of higher K-groups at the time [3, §0]. Segal's group completion of symmetric monoidal categories is in [7].

for a prime $\ell \neq p$ (for $\ell = p$ the positive-degree cohomology vanishes), obtained from $\mathbb{Z} \curvearrowright H^*(\mathrm{GL}(\overline{\mathbb{F}}_p), \mathbb{Z}/\ell)$ (Frobenius action).[7]

$$K(\mathbb{F}_p) \xrightarrow{\simeq} \mathbb{Z} \times BU^{h\mathbb{Z}},$$

where $\mathbb{Z} \curvearrowright BU$ via the "Frobenius map", i.e. the Adams operation ψ^p [5, Thm. 7].

Definition.

$$K(R) := BGL(R)^+ \times K_0(R) \simeq \big((\mathrm{proj}_R)^{\simeq}\big)^{gp}.$$

Here $(\mathrm{proj}_R)^{\simeq}$ is the groupoid of f.g. projective R-modules, a commutative monoid under $\oplus$, and $(-)^{gp}$ is its group completion; the equivalence is the group completion theorem.

2 K-theory of stable ∞-categories

Problems

- *Schemes:* there is no reasonable "thing" to group complete that gives a cohomology theory, and there is no M-V type sequence.[8]
- We want a long exact sequence

  $$\cdots \to \bigoplus K_0(\mathbb{F}_p) \longrightarrow K_0(\mathbb{Z}) \twoheadrightarrow K_0(\mathbb{Q}),$$

  but long exact sequences usually come from spectra:

  $$\bigoplus K(\mathbb{F}_p) \longrightarrow K(\mathbb{Z}) \longrightarrow K(\mathbb{Q}).$$

- *Group rings:* we want to study representation theory, but $K(k\pi)$ doesn't see a lot since there are very few f.g. projective $k\pi$-modules (when $\mathrm{char}\, k$ divides $|\pi|$). So they invented G-theory, the K-theory of all f.g. modules.
- *Waldhausen:* A-theory [9], $A(X) \simeq K(\mathbb{S}[\Omega X])$, and $A(X) \simeq K\big((\mathrm{Sp}_{/X})^{\omega}\big)$.

Observation (Waldhausen and many others)**.** $K(R)$ only depends on $\mathrm{Perf}(R)$ (or $D(R)$), the stable ∞-category of perfect complexes of R-modules.

Idea. Let $\mathcal{C}$ be a stable ∞-category. Set

$$K_0(\mathcal{C}) = \mathbb{Z}\{\text{equivalence classes of objects}\}\big/\big([x] + [z] = [y] \text{ for fiber sequences}\big).$$

Example.

1. The pushout square $\begin{array}{ccc} x & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \Sigma x \end{array}$ gives $[x] + [\Sigma x] = 0$.

[7] This is Quillen's computation [5]; it is the subject of the talks "Brauer lifts and $K(\mathbb{F}_p)$" and "Cohomology of $\mathrm{GL}_n(\mathbb{F}_q)$ I–II" of this seminar.

[8] For vector bundles on a scheme, short exact sequences need not split, so $\oplus$ alone gives the wrong relations. Quillen's Q-construction for exact categories [6] was the first fix.

2. The square $\begin{array}{ccc} \mathbb{Z} & \xrightarrow{p} & \mathbb{Z} \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{F}_p \end{array}$ gives $[\mathbb{F}_p] = 0$.

Situation $\rightsquigarrow$ stable ∞-category:

$$\begin{array}{lcl} R & \rightsquigarrow & \mathrm{Perf}(R) \\ \mathrm{Sch} \ni X & \rightsquigarrow & \mathrm{Perf}(X),\ D^b_{coh}(X) \\ \mathrm{An} \ni X & \rightsquigarrow & (\mathrm{Sp}_{/X})^\omega \\ G,\ k & \rightsquigarrow & \mathrm{Perf}(kG),\ \mathrm{Fun}(BG, \mathrm{Perf}\, k) \end{array}$$

Higher K-theory

From this perspective, take two maps $x \xrightarrow{f} y \xrightarrow{g} z$:

$$\begin{array}{ccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathrm{cofib}(f) & \longrightarrow & \mathrm{cofib}(gf) \\ & & \downarrow & & \downarrow \\ & & 0 & \longrightarrow & \mathrm{cofib}(g) \end{array}$$

Then

$$[z/y] + [y/x] + [x] = [z/x] + [x] = [z] = [z/y] + [y] = [z/y] + [y/x] + [x].$$

This is a loop!(?) The "higher coherences" give higher K-theory.

Remark. In the space $K(\mathcal{C})$, a cofiber sequence $a \to b \to c$ does not give an equality $[b] = [a] + [c]$ but a *path* from $[a] + [c]$ to $[b]$. The four equalities above use four cofiber sequences, read off from the diagram:

$$\begin{array}{rl} [z/y] + [y/x] + [x] = [z/x] + [x] & \text{from } y/x \to z/x \to z/y \text{ (right column)}, \\ [z/x] + [x] = [z] & \text{from } x \to z \to z/x, \\ [z] = [z/y] + [y] & \text{from } y \to z \to z/y, \\ [z/y] + [y] = [z/y] + [y/x] + [x] & \text{from } x \to y \to y/x. \end{array}$$

Each is a path, and the last one ends where the first one starts. Composing them gives a closed path, i.e. a loop in $K(\mathcal{C})$ based at $[x] + [y/x] + [z/y]$, so possibly an element of $\pi_1 K(\mathcal{C}) = K_1(\mathcal{C})$.

This particular loop is contractible: $K(\mathcal{C})$ is built so that the whole diagram above (the maps $x \to y \to z$ together with all their cofibers) gives a 2-dimensional disc whose boundary is exactly these four paths. Such discs are the "higher coherences". Other loops are not contractible: an automorphism α of x gives a cofiber sequence $x \xrightarrow{\alpha} x \to 0$, hence a path from $[x]$ to $[x]$, i.e. a loop, and its class in $K_1(\mathcal{C})$ is the element $(x, [\alpha])$ from the description of K_1 above.

Theorem. *For a ring R, $K(\mathrm{Perf}(R)) \simeq K^{\oplus}(R)$, the usual one.*[9]

Theorem (Localization)**.** *For (nice) null-sequences of stable ∞-categories $\mathcal{A} \to \mathcal{B} \to \mathcal{C}$, the sequence*

$$K(\mathcal{A}) \to K(\mathcal{B}) \to K(\mathcal{C})$$

is a fiber sequence.[10]

[9] Gillet–Waldhausen theorem; cf. [8, §1.11].

[10] "nice" means $\mathcal{A} \to \mathcal{B}$ is fully faithful and $\mathcal{C}$ is the Verdier quotient $\mathcal{B}/\mathcal{A}$ (up to idempotent completion, where one needs non-connective K-theory); see [8, 2].

Example.

$$\mathrm{Perf}(\mathbb{Z})^{p^\infty\text{-tors}} \longrightarrow \mathrm{Perf}(\mathbb{Z}) \longrightarrow \mathrm{Perf}(\mathbb{Z}[\tfrac{1}{p}]).$$

Every object of $\mathrm{Perf}(\mathbb{Z})^{p^\infty\text{-tors}}$ is equivalent to a finite direct sum $\bigoplus_i \Sigma^{n_i}\mathbb{Z}/p^{k_i}$, since over $\mathbb{Z}$ every complex is equivalent to the sum of its shifted homology groups, and here these are finite abelian p-groups. So it suffices to compute the classes $[\Sigma^n\mathbb{Z}/p^k]$: the suspension Σ just gives a minus sign, and for $\mathbb{Z}/p^k$ the cofiber sequences $\mathbb{Z}/p \to \mathbb{Z}/p^k \to \mathbb{Z}/p^{k-1}$ give $[\mathbb{Z}/p^k] = k\,[\mathbb{Z}/p]$. Hence $[\Sigma^n\mathbb{Z}/p^k] = (-1)^n k\,[\mathbb{Z}/p]$ and $K_0(\mathrm{Perf}(\mathbb{Z})^{p^\infty\text{-tors}}) \cong \mathbb{Z} \cong K_0(\mathbb{F}_p)$. By the next theorem even $K(\mathrm{Perf}(\mathbb{Z})^{p^\infty\text{-tors}}) \simeq K(\mathbb{F}_p)$, so localization gives the fiber sequence $K(\mathbb{F}_p) \to K(\mathbb{Z}) \to K(\mathbb{Z}[\frac{1}{p}])$ asked for in the Problems.

Besides localization, there are a few other main theorems to work with K-theory:

Theorem (Heart, Dévissage)**.** *For certain $\mathcal{A}$'s with extra structure, $K(\mathcal{A})$ is invariant under underiving and nil-thickening.*[11]

3 K-theory today

structural side $\rightleftarrows$ computational side

Structural side

- stable categories
- polynomial stuff
- organizing principles (motives)

Computational side

- trace methods
- prismatic, syntomic cohomology

References

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[11] *Underiving* is the theorem of the heart [1]: if $\mathcal{C}$ has a bounded t-structure, then $K(\mathcal{C}) \simeq K(\mathcal{C}^\heartsuit)$, the K-theory of the abelian category in its heart. *Nil-thickening* is Quillen's dévissage [6]: if every object of an abelian category $\mathcal{A}$ has a finite filtration with subquotients in a full abelian subcategory $\mathcal{B}$ (closed under subobjects and quotients), then $K(\mathcal{B}) \simeq K(\mathcal{A})$. Example: $\mathrm{Perf}(\mathbb{Z})^{p^\infty\text{-tors}}$ has heart the finite abelian p-groups, and a finite abelian p-group M is filtered by $M \supset pM \supset p^2M \supset \cdots \supset 0$ with $\mathbb{F}_p$-vector spaces as subquotients, so $K(\mathrm{Perf}(\mathbb{Z})^{p^\infty\text{-tors}}) \simeq K(\text{finite abelian } p\text{-groups}) \simeq K(\mathbb{F}_p)$.

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